

Muhammed Ali Mehmood

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EDUCATION

PhD Mathematics, Imperial College London

Oct. 2022 – Mar. 2026

- **Key themes:** Machine Learning, Partial Differential Equations, Mathematical Modelling.
- **Five research papers, including:**
 - * "*Randomised Neural Networks for non-linear PDEs*" (in progress), L. Gonon and M.A. Mehmood.
 - * "*Duality solutions to the hard-congestion model for the dissipative Aw-Rascle system*", N. Chaudhuri, M.A. Mehmood, C. Perrin and E. Zatorska, Communications in Partial Differential Equations (2024)

MSc Pure Mathematics, Imperial College London

Oct. 2021 – Oct. 2022

- **Grade:** First Class Honours (79%) completing modules such as Probability Theory and Formalising Mathematics.

BSc Mathematics, King's College London

Sept. 2018 – Aug. 2021

- **Grade:** First Class Honours (83%) completing modules such as Numerical & Computational Methods with Python, Statistical Inference, Linear Partial Differential Equations, Fundamentals of Probability Theory

QUANT EXPERIENCE

Spring into Quant Finance Participant, G-Research

Apr 2025

- Placed on a highly competitive week-long Quant Finance program hosted by G-Research. The week provided a detailed introduction and immersion into the world of quantitative finance and machine learning.
- Workshops included understanding and implementing gradient boosting models (e.g. XGBoost) and an introduction to quant finance and financial markets.

ML PROJECTS

LLMs for Exam Question Classification (m6ths.com)

Mar 2025 – Current

- Deployed and fine-tuned a BERT-based multi-label model on 1 800 + tagged A-Level Mathematics questions, achieving 96% accuracy / 93 % precision & recall on unseen 2024 papers.
- Automated the workflow: PDFs → individual-questions → OCR → topic prediction → topic-specific PDF bundles; used this to generate a large question bank of A-Level Math exam questions hosted at **m6ths.com** (in alpha).

Machine Learning Algorithms from Scratch - Educational Notebooks

Oct 2024

- Built a set of Jupyter notebooks implementing core ML algorithms (e.g. k-NN, Naive Bayes, decision trees, neural networks) from first principles using only Numpy.
- Designed as a pedagogical resource showcasing deep understanding of ML fundamentals and model mechanics.

AWARDS

- **Doris-Chen Award (2024)** - an award of 1000 GBP given to exceptional PhD students in the Mathematics department of Imperial College.
- **Roth Scholarship (2022)** - awarded a full scholarship to undertake my PhD research at Imperial College.
- **MSc Award (2021)** - awarded the prize for the best MSc Pure Mathematics presentation at Imperial College.

SKILLS

- **Programming:** Python, Lean, C++ (Basic), $\text{\LaTeX}$
- **Software:** NumPy, pandas, PyTorch, scikit-learn